

Peptic Ulcer Disease: Drug Treatment

High-yield revision · YMnotes Digestive System pp. 44–49

Name clues: PPI **-prazole** · P-CAB **-prazan** · H2 blocker **-tidine**

1 Reduce acid production

ANTI-SECRETORY DRUGS

Proton pump inhibitors (PPIs) **-prazole**

Omeprazole, esomeprazole, lansoprazole, pantoprazole

- **Prodrug**, activated by acid → **irreversibly** blocks H^+/K^+ ATPase
- Blocks the **final step** of HCl secretion → **most effective** acid suppressant
- Enteric-coated; take **30–60 min before meals**
- $t_{1/2}$ 1 h but acts **24 h** → once daily
- **Lansoprazole**: strongest anti-*H. pylori* effect
- ADRs: infections (pneumonia), **vit B12 deficiency**, ↓ activation of **clopidogrel**

H2-receptor antagonists **-tidine**

Cimetidine (prototype), famotidine, nizatidine

- Reversible H2 block on parietal cells
- Best against **nocturnal / fasting** acid
- **Cimetidine**: enzyme inhibitor, antiandrogenic, CNS **effects** (gynecomastia, impotence, confusion in elderly)
- Tolerance & **rebound hyperacidity**
- Rapid IV → **bradycardia & hypotension**
- Nizatidine = most potent

K⁺-competitive acid blockers (P-CABs) **-prazan**: vonoprazan

Newer alternative to PPIs: **reversible, not a prodrug, acid-stable** (take with or without food), full effect in **1 day** (PPI 3–5 days).

Uses: peptic ulcer · *H. pylori* regimens · GERD · Zollinger-Ellison (high-dose PPI) · stress ulcer · before C-section (prevent aspiration). PPIs beat H2 blockers in efficacy and once-daily dosing.

2 Protect the lining

MUCOSAL PROTECTIVES

	How it works	Main use	Watch out
Sucralfate	Needs acid → sticky barrier over ulcer base	Stress-ulcer prophylaxis	Give antacids / H2 blockers 1–2 h apart ; constipation
Misoprostol	PGE1 analogue : ↓ acid, ↑ mucus & HCO_3^-	NSAID-induced ulcer	Diarrhea; contraindicated in pregnancy (abortion)
Bismuth	Kills <i>H. pylori</i> + coats ulcer	Quadruple therapy	Black tongue & stools (mimic bleeding); max 2 months

3 Eradicate *H. pylori*

COMBINATION THERAPY

Eradication = **rapid healing & fewer recurrences**. Remember **P-CAT**: PPI + Clarithromycin + Amoxicillin + Tinidazole/metronidazole.

Triple therapy · 10–14 days

- PPI + **clarithromycin** + **amoxicillin**
- Amoxicillin allergy → use **metronidazole**

Quadruple therapy · 10–14 days

- PPI + **bismuth** + **metronidazole** + **tetracycline** (4× daily)
- Other options: sequential, levofloxacin-based

Then **PPI for 4–6 weeks** to heal the ulcer. A P-CAB can replace the PPI.

4 Neutralise existing acid

ANTACIDS

- **Weak bases** that **neutralise** HCl → **rapid** but short relief (1–2 h) of ulcer pain, dyspepsia, GERD
- Taken **1 and 3 h after meals and at bedtime**

Antacid	Key side effects
$\text{Al}(\text{OH})_3$	Constipation · chelates phosphate → osteomalacia, encephalopathy (renal patients)
Mg^{2+} salts	Diarrhea · CNS depression (renal patients)
CaCO_3	Rebound hyperacidity · milk-alkali syndrome (alkalosis + hypercalcemia)
NaHCO_3 (systemic)	Systemic alkalosis · Na^+ → salt & water retention (avoid in HF, hypertension)

Best choice: Al + Mg together (e.g. magaldrate): they **cancel each other's bowel effects** with no systemic alkalosis.

Interaction: Al^{3+} chelates tetracyclines, quinolones, digoxin, iron → give **2 h apart**.

Top 10 for the exam

1. Most effective acid suppressant = **PPI**
2. PPI before meals; P-CAB any time
3. Long-term PPI → **B12 deficiency**
4. Cimetidine → **gynecomastia**, drug interactions
5. H₂ blockers best for **night-time** acid
6. Misoprostol: NSAID ulcer, **not in pregnancy**
7. Bismuth → **black stools**
8. Al = constipation, Mg = diarrhea → combine
9. CaCO_3 → **milk-alkali syndrome**
10. Antacids: **2 h gap** from tetracyclines & quinolones